

Umesh Sharma

Bangalore, India

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SUMMARY

AI/ML-focused Data Science undergraduate skilled in Python, Machine Learning, and LLM-based systems. Experienced in building end-to-end AI applications including RAG pipelines, automation workflows, and scalable backend systems. Strong interest in Applied AI, GenAI, and real-world problem solving.

EDUCATION

B.S. (Honours) in Data Science & Artificial Intelligence <i>Indian Institute of Technology Madras (IITM)</i>	2024 – 2028
UG in Computer Science <i>PW Institute of Innovation</i>	2023 – 2027

EXPERIENCE

AI/ML Intern <i>PW LeapX</i>	Dec 2025 – Present <i>Bangalore, India</i>
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- Developing Applied AI systems including LLM-powered applications and end-to-end ML pipelines
- Building scalable backend services using FastAPI with modular architecture, REST APIs, and authentication
- Working on real-world NLP and automation use cases integrating AI models into production workflows
- Collaborating to design, test, and optimize production-ready AI solutions

TECHNICAL SKILLS

- **Programming:** Python, SQL, C (basic)
- **AI/ML:** Regression, Classification, Supervised & Unsupervised Learning
- **LLM/GenAI:** RAG, Prompt Engineering, LLM Integration
- **Libraries:** NumPy, Pandas, Scikit-learn, NLTK, Matplotlib, Seaborn
- **Frameworks/Tools:** FastAPI, Streamlit, n8n, Docker, Git
- **Cloud/Infra:** GCP, AWS S3
- **Databases:** PostgreSQL

PROJECTS

- **Souli — AI-Powered Inner Wellness Voice Companion**
Python, Qdrant, Ollama, Whisper, sentence-transformers
 - Built an emotionally aware conversational AI system using a multi-phase state machine for structured dialogue flow
 - Designed a Retrieval-Augmented Generation (RAG) pipeline using Qdrant and sentence-transformers for semantic retrieval from counseling data
 - Implemented embedding-based classification to map user inputs into predefined emotional states (energy nodes)
 - Developed real-time speech understanding pipeline using Whisper for transcription and LLM-based response generation
- **CallAudit – AI-Based Call Quality Analysis System**
Python, FastAPI, Azure OpenAI, PostgreSQL
 - Built an LLM-powered system for automated call auditing and quality evaluation from transcribed conversations
 - Integrated Azure OpenAI to analyze call transcripts and generate structured quality insights
 - Designed pipelines for processing and analyzing conversational data using NLP techniques

- Implemented role-based analysis logic to support different evaluation perspectives

- **AI-Powered Email Assistant Workflow**

n8n, OpenAI (GPT-4.1-mini)

- Developed an AI-driven email processing system for automated classification and response generation
- Used LLMs to extract sender information and perform semantic categorization of email content
- Implemented conditional logic for personalized response generation based on extracted context

- **AI Personal Assistant using LLM Agents**

n8n, Groq (LLaMA 4)

- Built an LLM-based personal assistant capable of automating multi-step tasks using structured tool-calling
- Designed prompt engineering strategies for reliable task execution and intent handling
- Implemented deterministic execution logic to reduce ambiguity and improve automation accuracy

CERTIFICATIONS

- Machine Learning Bootcamp – Udemy
- SQL Certification – Simplilearn